Tagging and releases

A SWAT+ version is a Git tag. No source file is edited to make a release: the build reads the tag and writes it into the executable's name and its start-up banner.

Version numbers

Current tags have three parts, revision, minor and patch, for example `62.0.0` and `62.0.1`. Older tags include `61.0`, `61.0.2.61` and `60.5.7`. List them with:

```
git tag --sort=-creatordate | head
```

At configure time CMake runs `git describe --tags`:

- on a tagged commit the version is the tag, for example `62.0.1`;
- on a later commit it is the tag, the number of commits since, and the abbreviated commit name, for example `62.0.1-3-g1a2b3c4`;
- outside a Git clone it is `unknown`.

A version can also be given by hand with `-D TAG=<version>` when configuring.

Version in the executable and outputs

The executable name, `swatplus-<version>-<compiler>-<system>_<arch>[-Rel|-Dbg]`, for example `swatplus-62.0.1-gnu-l`

The start-up banner, printed to the screen and to `simulation.out`:

```
                    SWAT+
              Revision 62.0.1
      Soil & Water Assessment Tool
GNU (13.3.0), 2026-10-19 22:51:37, Linux
    Program reading . . . executing
```

The title line of every output file, which contains the build date and `Rev <year>.<version>`:

```
 demo                        SWAT+ 2026-10-19        MODULAR Rev 2026.62.0.1
```

Building a tagged version

```
git checkout 62.0.1
cmake -B build -D CMAKE_BUILD_TYPE=Release
cmake --build build -j 8
```

A tag on GitHub can be moved after it is first published (the 62.0.1 tag was replaced on 19 October 2026). Run `git fetch --tags --force` before checking out a tag in an existing clone, so that the local tag matches the one on GitHub.

Making a release

1. Merge the changes for the release into `main`.
2. Tag the merge commit and push the tag:

   ```
   git tag 62.0.2
   git push origin 62.0.2
   ```

3. Pushing a tag of the form `x.y` or `x.y.z` starts the release workflow, which builds the executables for Windows, Linux and macOS with gfortran and the Intel compilers and attaches the zipped executables to a GitHub release.
4. Write the release notes on the GitHub release page, and archive the release on Zenodo (see Citing and license).
5. Tag the documentation repository with the same version, which publishes a matching version of this site.

Sources

- CMakeLists.txt: `if(NOT TAG)` block with `git describe --tags`, `SWATPLUS_EXE`, and `configure_file` of `src/main.f90.in` with `ISO`, `TODAY` and `YEAR` time stamps.
- src/main.f90.in: `prog = " SWAT+ @TODAY@ MODULAR Rev @YEAR@.@SWAT_VERSION@"` and the banner format. The generated `src/main.f90` is overwritten on every configure; edit `main.f90.in`.
- doc/Tagging.md.
- .github/workflows/build.yml: runs on tags matching `*.*` and `*.*.*`.
- Banner and title line: output of the 62.0.1 `gnu-lin_x86_64` release executable.